Section B (20 marks)

7 **(a)** **(i)** State and explain the effect of the width of aperture on the diffraction of the wave.

……

……

……

……[2]

(ii) The band is practising in the band room down the corridor with the door left slightly ajar. Explain why notes of a certain range of frequencies can be heard more clearly than others at the other end of the corridor.

……

……

……

……[2]

(b) A plane diffraction grating is illuminated normally by a monochromatic light source of wavelength λ. Light rays from each slit travel a different distance before arriving at a particular point on the screen. In order to derive an expression for the path difference, the light rays that meet at the same point on the screen can be approximated as parallel rays.

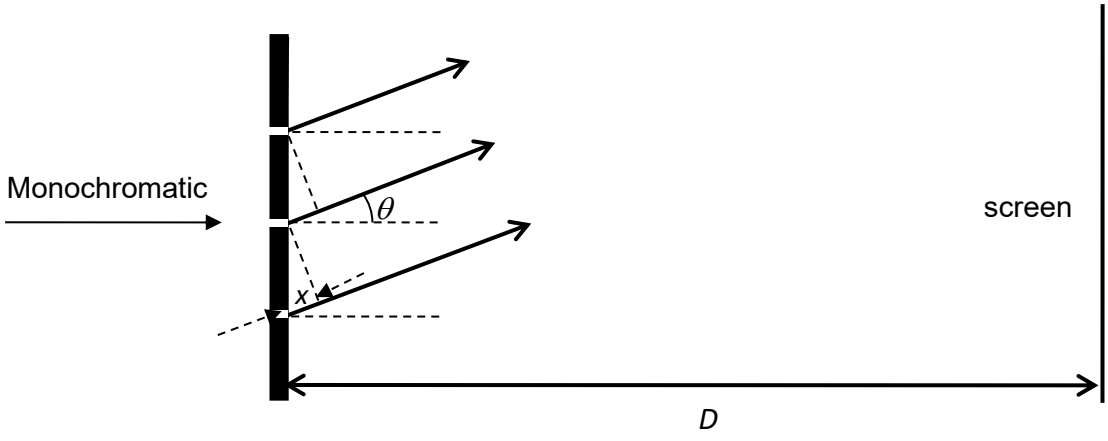

Fig. 7.1

(i) State the condition where the rays can be assumed to be parallel.

……

……[1]

7 **(b)** **(ii)** Derive an expression for the path difference x between adjacent rays in terms of p and θ, where p is the number of lines per unit length of the grating, and θ is the angle between the emergent ray and the normal. [2]

(iii) Hence, derive an expression relating p and θ for the constructive interference of the rays. Define any additional symbol you use. [2]

7 **(c)** A diffraction grating is set up at the centre of a rotating table which completes a revolution every 3.0 s. The grating is illuminated normally by monochromatic light of wavelength λ from a source which is also mounted on the table, see Fig. 7.2.

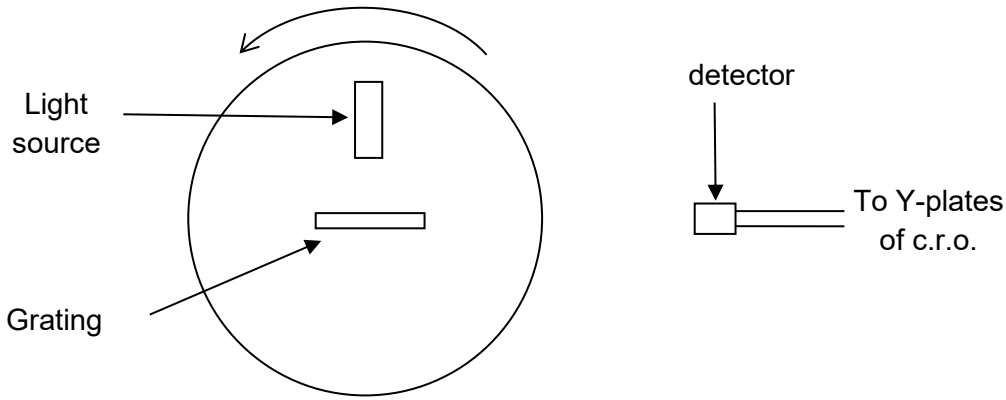

Fig. 7.2

The emergent beams of light from the grating are monitored by means of a stationary opto-electrical detector. The output from detector is displayed on a cathode ray oscilloscope (c.r.o.). With the time-base set at 0.10 s cm^{-1}, the trace obtained is shown in Fig. 7.3. The relative positions of the peaks are as indicated.

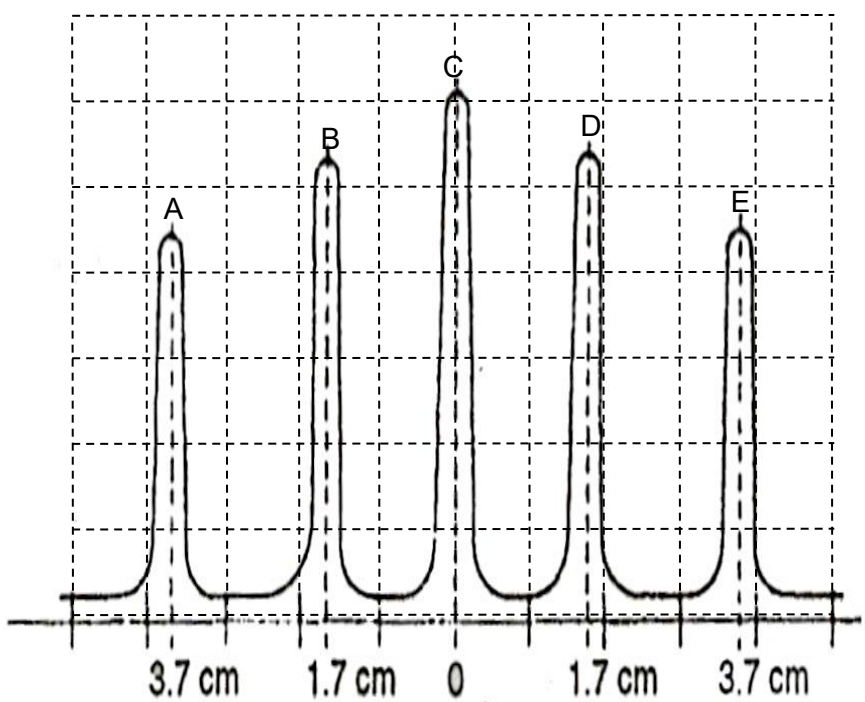

Fig. 7.3

7 **(c)** **(i)** Calculate the angular speed of rotation of the grating.

angular speed = …….rad s^{-1} [1]

(ii) Explain the appearance of the trace in Fig. 7.3.

……..……

……..……

……..……

…….[2]

(iii) Calculate the angle θ in radians, as defined in (b) (ii), for peaks D and E.

θ for D = …….radians

θ for E = …… radians

[3]

7 **(c)** **(iv)** Using peak E, calculate the wavelength of the light if the grating has 550 lines per mm.

wavelength = ……nm [2]

(v) Explain why it is preferable to calculate the wavelength using peak E rather than peak D.

……

……[1]

(vi) Sketch in Fig. 7.4, the c.r.o. display, if the diffraction grating is replaced by a double slit of the same slit separation as the diffraction grating. [2]

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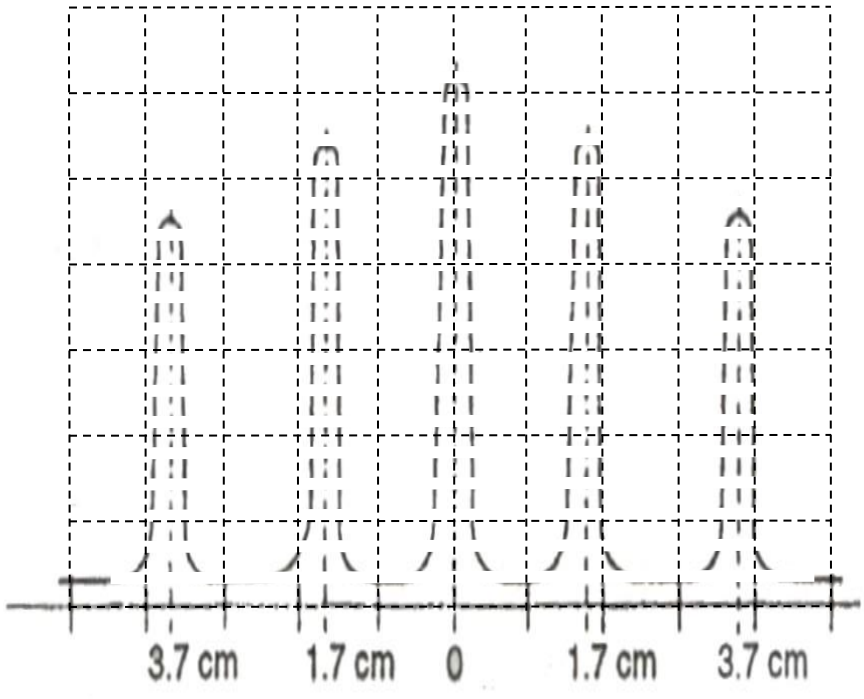

Fig. 7.4

(The original display from the grating is shown in dotted line)